Open-Source Firewalls

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OPNsense

- Runs on FreeBSD
- Fork of pfSense
- Includes DDOS protection, IPS & IDS and Packet Filter
- Less abstractions than the other two firewalls



IPFire

- Fork of IPCop
- Uses iptables for rule setting
- Web GUI interface

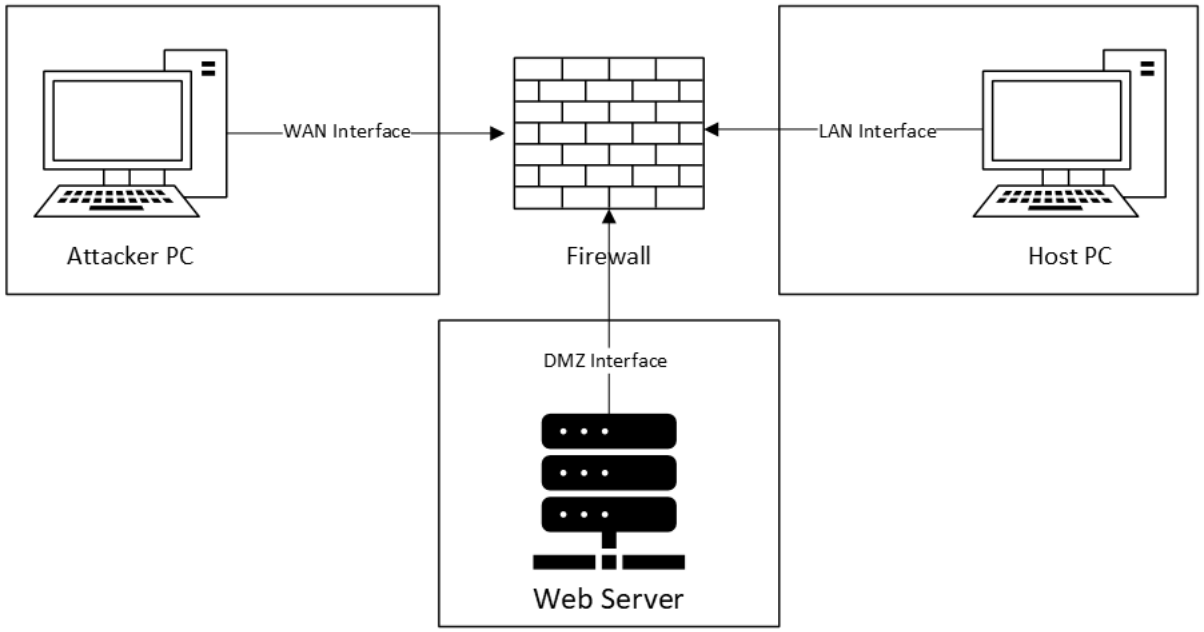


IPCop

- Fork of Smoothwall
- Obsolete
- Simplistic web interface



Test topology



Tests performed

- Reconnaissance attacks using Nmap
- Denial of service attacks (ICMP & HTTP) using hping3 and WRK
- Brute forcing of SSH

Reconnissance attacks

- nmap –A –p- <IP-Address>

```
└─$ nmap -A -p- 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-27 12:37 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl
ed. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.103
Host is up (0.00044s latency).
Not shown: 65532 filtered tcp ports (no-response), 1 filtered tcp ports (port
-unreach)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.0 (protocol 2.0)
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.58 (Ubuntu)
MAC Address: 08:00:27:05:58:1C (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Device type: general purpose|firewall|storage-misc
Running (JUST GUESSING): Linux 4.X|5.X|6.X|2.6.X|3.X (95%), IPFire 2.X (90%),
 QNAP QTS 5.X (90%)
OS CPE: cpe:/o:linux:linux_kernel:4.19 cpe:/o:linux:linux_kernel:5 cpe:/o:ipf
ire:ipfire:2.25 cpe:/o:linux:linux_kernel:6.1 cpe:/o:qnap:qts:5.0 cpe:/o:linu
x:linux_kernel:5.10 cpe:/o:linux:linux_kernel:2.6 cpe:/o:linux:linux_kernel:3
Aggressive OS guesses: Linux 4.19 (95%), Linux 4.15 (91%), Linux 5.10 - 5.19
(91%), IPFire 2.25 firewall (Linux 4.14) (90%), IPFire 2.27 (Linux 5.15 - 6.1
) (90%), QNAP QTS 5.0 (Linux 5.10) (90%), Linux 2.6.32 - 3.0 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   0.44 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 133.07 seconds
```

Reconnissance attacks

- nmap –A –p- <IP-Address>

```
┌──(kali㉿kali)-[~]
└─$ nmap -A -p- 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-28 15:58 EDT
Nmap scan report for 192.168.56.103
Host is up (0.00099s latency).
Not shown: 65534 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 08:00:27:05:58:1C (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Device type: general purpose|router|firewall
Running (JUST GUESSING): Linux 4.X|5.X|2.6.X|3.X (97%), MikroTik RouterOS 7.X
 (89%), IPFire 2.X (87%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:
linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:3.10 cpe:/o:linux:linux_kernel:
6.0 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3 cpe:/o:ipfire:
ipfire:2.27
Aggressive OS guesses: Linux 4.19 - 5.15 (97%), Linux 4.15 - 5.19 (91%), Linu
x 5.0 - 5.14 (91%), OpenWrt 21.02 (Linux 5.4) (91%), Linux 2.6.32 (90%), Linu
x 2.6.32 or 3.10 (90%), Linux 4.19 (89%), Linux 6.0 (89%), MikroTik RouterOS
7.2 - 7.5 (Linux 5.6.3) (89%), Linux 4.0 - 4.4 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   1.00 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 168.84 seconds
```

Denial of service attacks

```
└─$ sudo hping3  -1 --flood -f -d 1500 192.168.183.243
```

```
└─$ sudo hping3  -1 --flood --rand-source -x -d 1500 192.168.183.243
```

```
└─$ wrk -t 4 -c 4 -d 20 http://192.168.183.243
Running 20s test @ http://192.168.183.243
  4 threads and 4 connections
  Thread Stats   Avg      Stdev     Max   +/- Stdev
    Latency     2.01ms  544.32us  11.97ms   79.84%
    Req/Sec   498.17     47.39   650.00     82.50%
  39730 requests in 20.04s, 413.99MB read
Requests/sec:   1982.37
Transfer/sec:     20.66MB
```

Before

```
┌──(kali㉿kali)-[~]
└─$ wrk -t 4 -c 4 -d 20 http://192.168.183.243
Running 20s test @ http://192.168.183.243
  4 threads and 4 connections
  Thread Stats   Avg      Stdev     Max   +/- Stdev
    Latency    68.47ms   15.60ms 131.09ms   83.63%
    Req/Sec    14.59      5.16    20.00     49.87%
  1161 requests in 20.09s, 12.10MB read
Requests/sec:     57.78
Transfer/sec:    616.57KB
```

After

SSH Attacks

Simple SSH login attempts:

Before

```
┌──(kali㉿kali)-[~]
└─$ ssh -p 8022 root@192.168.56.101
root@192.168.56.101's password:
Permission denied, please try again.
root@192.168.56.101's password:
Permission denied, please try again.
root@192.168.56.101's password:
root@192.168.56.101: Permission denied (publickey,password,keyboard-interactive).

┌──(kali㉿kali)-[~]
└─$ ssh -p 8022 root@192.168.56.101
root@192.168.56.101's password:
Permission denied, please try again.
root@192.168.56.101's password:
Permission denied, please try again.
root@192.168.56.101's password:
root@192.168.56.101: Permission denied (publickey,password,keyboard-interactive).

┌──(kali㉿kali)-[~]
└─$ ssh -p 8022 root@192.168.56.101
root@192.168.56.101's password:
Permission denied, please try again.
root@192.168.56.101's password:
Permission denied, please try again.
root@192.168.56.101's password:
root@192.168.56.101: Permission denied (publickey,password,keyboard-interactive).

┌──(kali㉿kali)-[~]
└─$ ssh -p 8022 root@192.168.56.101
root@192.168.56.101's password:
Permission denied, please try again.
root@192.168.56.101's password:
Permission denied, please try again.
root@192.168.56.101's password:
root@192.168.56.101: Permission denied (publickey,password,keyboard-interactive).

┌──(kali㉿kali)-[~]
└─$ ssh -p 8022 root@192.168.56.101
root@192.168.56.101's password:
```

After

```
┌──(kali㉿kali)-[~]
└─$ ssh -p 8022 root@192.168.56.101
ssh: connect to host 192.168.56.101 port 8022: Connection timed out
```

Brute forcing (hydra):

```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -t 3 -P /usr/share/wordlists/rockyou.txt.gz ssh://192.168.56.101:8022
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service
 organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-06 19:39:30

[DATA] max 3 tasks per 1 server, overall 3 tasks, 14344399 login tries (l:1/p:14344399), ~4781467 tries
per task
[DATA] attacking ssh://192.168.56.101:8022/
[8022][ssh] host: 192.168.56.101   login: root   password: password
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-06 19:39:32
```

```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -t 3 -P /usr/share/wordlists/rockyou.txt.gz ssh://192.168.56.101:8022
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service
 organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-06 19:47:44
[DATA] max 3 tasks per 1 server, overall 3 tasks, 14344399 login tries (l:1/p:14344399), ~4781467 tries
per task
[DATA] attacking ssh://192.168.56.101:8022/
[STATUS] 3947.00 tries/min, 3947 tries in 00:01h, 14340452 to do in 60:34h, 3 active
[8022][ssh] host: 192.168.56.101   login: root   password: dublin
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-06 19:48:48
```

```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -t 3 -P /usr/share/wordlists/rockyou.txt.gz ssh://192.168.56.101:8022
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service
 organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-07 07:12:58
[DATA] max 3 tasks per 1 server, overall 3 tasks, 14344399 login tries (l:1/p:14344399), ~4781467 tries
per task
[DATA] attacking ssh://192.168.56.101:8022/
[ERROR] could not connect to ssh://192.168.56.101:8022 - Timeout connecting to 192.168.56.101
```

Results

Attack	OPNsense	IPCop	IPFire
Reconnissance	Partially mitigated	Partially mitigated	Mitigated
ICMP Flood	Not mitigated	Partially mitigated	Mitigated
HTTP Flood	Mitigated	Mitigated	Mitigated
SSH Attack	Mitigated	Partially mitigated	Mitigated